



An experimental comparison of deep learning models for pneumonia classification from chest X-ray images

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ABSTRACT

Chest X-rays are valuable for diagnosing and monitoring various conditions such as pneumonia, COVID-19, and heart disease, to cite but a few. This study specifically focuses on X-ray image classification between non-pathological lung images (i.e., normal/healthy) and lung images affected by pneumonia. The latter category is further subdivided into common pneumonia (bacterial and viral) and COVID-19 pneumonia. In this paper, we experimentally compare state-of-the-art machine learning methods tackling this multi-class classification problem. Specifically, we design a comparative methodology to put to test six prominent Convolutional Neural Network architectures, including *ResNet*, *AlexNet*, *VGGNet*, *SqueezeNet*, *DenseNet*, and *InceptionV3*. Our work overcomes the issue that the existing proposals for tackling pneumonia classification based on Deep Learning are not directly comparable, due to considerable differences in their respective modeling pipeline. In this paper, we conduct a comparative evaluation based on a uniform pipeline, from data collection, curation, and all the way to model validation. We introduce a balanced, labeled dataset, including 6,939 X-ray images. In pursuit of high accuracy, we also integrate (in all six cases) the same data curation, transformation, and augmentation techniques. Key findings demonstrate *VGGNet*'s superior performance, with an overall 97% accuracy, making it highly applicable to real-world clinical scenarios.

1. Introduction

1.1. Research context and motivation

The World Health Organization (WHO) asserts that, every year, 1.4 million children in the 0–5 age range die from pneumonia, accounting for 18% of all deaths of children under five years old worldwide. Pneumonia is among the top ten causes of death in the United States [1]. In India, more than 3700 children die annually because of it, which accounts for 50% of all deaths due to this disease [2]. The WHO reported that between January 2020 and May 2022, Saudi Arabia alone had 759,856 confirmed COVID-19 cases, including 9118 deaths. The highest concentration of individuals affected by pneumonia was found in South Asia and sub-Saharan Africa [3].

There is also a broad variation of cases, across different categories of people. Those at higher risk of contracting pneumonia are: (i) people over 65 years old, (ii) smokers, (iii) people having a weak immune systems, and (iv) children, especially those who are younger than five years old [4]. It is also important to mention that the type of medical

treatment is highly dependent upon the type of pneumonia, its severity, and the general health condition of the patients affected by this disease. It is therefore, crucial to develop robust AI techniques to aid the fast and reliable diagnosis of pneumonia, which translates into a machine learning problem known as a multi-class classification.

Generally speaking, an effective process for diagnosing chest diseases starts from chest X-rays, which are broadly accessible at low cost. These images may be used to determine if there is fluid in or around the lungs, or to evaluate the air surrounding a lung. For instance, doctors may prescribe a chest X-ray if a patient gets hurt, has a chest injury, or has shortness of breath. Doctors may also use chest X-rays to monitor the progress of a specific disease. For example, X-rays help to determine whether a patient has bacterial or viral pneumonia (such as COVID-19), heart problems, cancer, or other medical conditions.

Specialist physicians are trained to diagnose a range of pathologies directly from X-rays. Misdiagnosis is, however, also possible due to fatigue, distraction, and generally human error. Other errors may be incurred by trainees or junior physicians who may not always be

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mentored by more experienced radiologists. For all these cases, it is valuable to employ Machine Learning (ML) classifiers alongside the X-rays to aid (and accelerate) the diagnostic process and, possibly, reduce misdiagnosis, particularly when hospitals face shortage of experienced radiologists.

Detecting pneumonia and, specifically, being able to differentiate among different types of pneumonia (e.g. bacterial, viral, COVID-19), is therefore a crucial task. Any means for improving the automation of the diagnostic process, must therefore be highly accurate and reliable, since people's lives are at stake, in face of reduced resources available in the healthcare systems [5].

Our work is developed in this medical context, aiming to experimentally evaluate the state-of-the-art in medical computer vision based on Deep Learning (DL). We aim to establish how well DL may perform in the multi-classification task that differentiates among healthy, pneumonia or COVID-19 conditions, directly from a standard chest X-ray. We explore and experimentally compare the effectiveness of several DL architectures, considering a completely automated process, not considering human intervention. However, in real-life applications, DL would be used in tandem with humans to achieve even higher confidence in the diagnosis. We should stress that all DL models scrutinized herein can potentially scale up in a real-world context, with significant implications in both diagnostic accuracy and cost efficiency.

We should note that this DL classification problem has been extensively addressed in recent literature, particularly due to the COVID-19 pandemic and the significant developments in Machine Learning (ML), Deep Learning (DL), and computer vision - notable examples are found in [6–10]. Nevertheless, it is hardly possible to fairly determine how each approach compares to the others, due to significant differences in their respective ML pipelines. These differences include data availability, dataset types, data curation, pre-processing and transformation methods, neural network architectures, and tuning and validation processes.

Herein, we provide a rigorous and uniform data science pipeline, from data ingestion to model validation, to conduct a fair comparison among prominent CNN classification models, including *i.e.*, *ResNet*, *AlexNet*, *VGGNet*, *SqueezeNet*, *DenseNet*, and *InceptionV3*.

In our study we focus on detecting three classes: (i) non-pathological (*i.e.*, healthy lungs), (ii) general pneumonia (both bacterial and viral), and (iii) pneumonia caused by COVID-19.

1.2. Deep learning strengths and limitations on medical applications

Since the earliest development of computers, there has always been an interest in utilizing these systems to automate tasks, as computer programs are quick to address medical complexity. In addition, modern high-level programming languages, like Python, made it simple to develop and improve algorithms, for Deep Learning (DL) and artificial intelligence, capable of generating reliable predictions and classifications, given some raw input data like images.

Automated systems use both supervised learning - whereby models are trained on input-output pairs to learn explicit mappings - and unsupervised learning - whereby models uncover hidden patterns or structures from input data alone. Depending on the task, these systems may adopt DL techniques (e.g., Convolutional Neural Networks for image data or autoencoders for feature learning) as well as traditional Machine Learning (ML) methods (e.g., linear and logistic regression, decision trees, and kernel-based models).

Many ML methods, especially the traditional ones, share a common limitation, which is that they require specific features to be defined in advance. Feature extraction is often the most crucial step, as selecting appropriate features and their quantity can significantly affect prediction quality. On the other hand, DL methods mitigate this by learning hierarchical features directly from data, the choice of model architecture and training data still remains problem-dependent, making a universal *a priori* approach unfeasible.



Fig. 1. Sample images of the three types of X-ray plates considered in this study: pneumonia caused by COVID-19, healthy (labeled as Normal) lung, and common pneumonia, respectively.

However, ML models do not automatically guarantee a good solution, which emphasizes the importance of conducting comparative experimental studies, such as the one presented herein. One major limitation, especially in the field of DL, is overfitting, where a model performs exceptionally well on training data but poorly on testing data. This happens when the model memorizes the training data too closely, limiting its ability to generalize to new, unseen data (*i.e.*, test data).

Proper training requires significant time and expertise. For medical purposes, it is also crucial to handle sensitive data appropriately. While healthcare technologies can greatly enhance patient care, they raise concerns about patient privacy and the protection of health information. This issue can be addressed through effective de-identification, or anonymization, of patient data, ensuring compliance with the Health Insurance Portability and Accountability Act (HIPAA) to safeguard patient information.

In this context, we might also find Deep Convolutional Neural Networks (CNNs), a subset of the DL domain, which heavily depend on large and high-quality datasets to avoid overfitting.

Deep CNNs may offer solutions to Machine Learning limitations in the medical domain, potentially improving disease prediction accuracy, even outperforming the average radiologist. For instance, DL-based X-ray image diagnosis has shown to be highly beneficial, providing rapid and accurate diagnoses, comparable to those of expert radiologists [11].

The widespread use of Artificial Neural Networks (ANN) today stems from their ability to automatically learn input features from datasets, reducing the need for extensive human intervention and expertise in training the models.

1.3. Proposed experimental framework

Herein, we evaluate a range of Deep Convolutional Neural Networks (CNNs) for automated chest X-ray image classification, as these are effective AI methods in computer vision, image analysis, and feature detection. We considered six CNNs architectures to identify their effectiveness in the pneumonia classification problem at hand, including: (i) *ResNet*, (ii) *AlexNet*, (iii) *VGGNet*, (iv) *SqueezeNet*, (v) *DenseNet*, and (vi) *InceptionV3*. We trained, tuned, and evaluated classification models for each of these six methods, to detect three different conditions: (i) ‘*Pneumonia*’ that includes both bacterial and viral pneumonia, (ii) ‘*COVID-19*’, which represent the pneumonia caused by COVID-19, or (iii) ‘*Normal*’ that represents the healthy state (See samples in Fig. 1).

We conducted our analysis on a relatively larger dataset (*i.e.*, 6939 X-ray images) [12] compared with typical ones found in the literature (see Section 3). The curated dataset is available on the Kaggle repository¹ - and an expert radiologist screened the chest X-ray images to remove all unreadable or low-quality scans.

Because of the limited availability of patient data, we used Data Augmentation (DA). Data Augmentation is a data space solution to the

¹ Dataset “COVID19_Pneumonia_Normal_Chest_Xray_PA_Dataset” available at <https://www.kaggle.com/datasets/amanullahasraf/covid19-pneumonia-normal-chest-xray-pa-dataset>.

problem of limited data. DA techniques enhance the size and quality of training datasets for better DL models.

Herein, we employed a range of data augmentation techniques (including color space transformations, brightness adjustment, re-scaling, rotation, kernel filtering, image mixing, random erasing, feature space augmentation, and adversarial training) to improve class balance and reduce the risk of overfitting. Additionally, we applied model-level strategies such as transfer learning, batch normalization, and dropout, as detailed in Section 4.1.

Our analyses unveil that the *VGGNet* and *DenseNet* architectures were the most accurate methods, achieving overall classification accuracy of 94% and 91%, respectively.

2. Background material

Artificial Neural Networks (ANNs) provide a range of powerful techniques for finding patterns and analyzing data, such as Kohonen Self Organizing Neural Networks, Feed-forward Neural Networks, and Convolutional Neural Networks (CNNs) [13]. They are generally composed of input, hidden, and output layers. The input layer receives the input dataset, the hidden layer processes it, and the output layer gives the output that can be a classification or prediction, depending on the type of tasks at hand.

In our work, we use ANNs to recognize and detect chest disease patterns on given X-ray images. Among the different types of ANNs, CNNs are the most prominent ones for efficiently processing images, thanks to the convolutional and pooling layers that are able to extract relevant features from the images.

2.1. Convolutional neural networks

Currently, the most prominent Deep Learning architectures to deal with medical images are based on Convolutional Neural Networks (CNN). A CNN model contains several layers including small neuron collections, aimed at perceiving parts of the images. The results from all the parts in a layer partially overlap in such a way as to create a whole image representation, then repeating this process on subsequent X-ray images, allowing the system to learn from the image composition. This model differs from other ML methods for its excellent ability in detecting complex image-driven pattern recognition.

A CNN is typically composed by three layer types, namely convolution, pooling, and fully-connected, as shown in Fig. 2. The convolution and pooling layers are used for feature extraction, whereas the fully-connected ones map the extracted features to the final output classification. The CNN increases complexity with each layer, identifying more significant portions of the image. As the image data progresses through the CNNs' layers, the model recognizes more significant elements and shapes of the objects until it identifies the intended target [14].

In this paper, we focused on six distinct CNN architectures for image identification and classification tasks, including: *ResNet*, *AlexNet*, *VGGNet*, *SqueezeNet*, *DenseNet*, and *InceptionV3*.

2.2. Transfer learning for overcoming CNN limitations

Modern CNNs, like *ResNet*, *VGGNet*, and *DenseNet*, perform classification tasks accurately. However, these models are deep and complex. It is hard to train them without overfitting, as they need several images to learn the image classification model. A possible solution is to use Transfer Learning [15].

Transfer learning is defined as a reuse of a pre-trained model on a new problem. Indeed, transferring the knowledge of a pre-trained network on a larger dataset, and fine-tuning it for a specific job, is an effective technique to obtain high accuracy while reducing training time, particularly when the dataset is limited, as it allows to improve image recognition robustness.

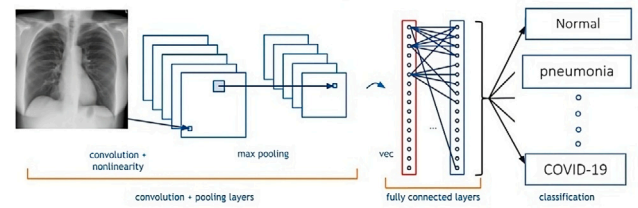


Fig. 2. The figure shows an example of the convolutional neural network (CNN) architecture used for chest X-ray image classification to highlights the process of feature extraction, pooling, and classification in medical image analysis using CNNs. On the left, an input X-ray image is fed into the network, which passes through multiple convolutional and max-pooling layers to extract features. These features are then flattened into a vector (vec) and passed through fully connected layers. The final output is a classification into one of three categories: Normal or healthy lungs, common pneumonia, or pneumonia caused by COVID-19.

Although disease classification tasks from chest X-ray images differ from traditional object classification tasks, these still share similar machine learning characteristics. Therefore, in our work model weights were initialized using pre-trained CNNs on a public image dataset, followed by fine-tuning for the specific task. During fine-tuning, some of the final layers were unfrozen, meaning their weights were made trainable and updated with each training step to adapt to the new data.

2.3. Performance metrics for classification

Herein, we use classification accuracy to measure the overall performance of the models under scrutiny. In addition, we are adopting the following evaluation methods, which provide greater comparative insights:

1. **Confusion Matrix:** A confusion matrix, also referred to as an error matrix, is a tabular representation of a classifier's performance, comparing actual class instances to predicted ones. It provides insights into the model's accuracy, precision, recall, and other evaluation metrics, and thus summarizes prediction results on a classification problem [16].

In the matrix, each row represents the true class and each column represents the predicted class. Diagonal elements indicate correct predictions for each class (true positives in binary classification), while off-diagonal elements indicate misclassifications (i.e., false positives and false negatives).

In this paper, the confusion matrix helped us to analyze model performance by highlighting correct and incorrect predictions across the three classes.

2. **Loss Function:** In an Artificial Neural Network (ANN), the loss function measures the difference between the actual and predicted outputs, quantifying the model's performance.

For our classification task, we used a Binary Cross-Entropy Loss Function, which returns a probability value between 0 and 1 and calculates the average difference between expected and predicted probabilities. When extending the model to classify multiple illnesses, we employed the Multi-class Cross-Entropy Loss Function for handling more than two classes [17].

3. Related work

3.1. Deep learning in general

Deep Learning-based methods are already being used in various domain areas such as healthcare, image captioning, Natural Language Processing, and fraud detection [18]. Several authors have proposed several biomedical image detection techniques. To cite some, M. I. Razaak

Table 1

Medical applications overview of DL techniques in literature. The table summarizes methods, advantages, and limitations.

Reference	Method	Pros	Cons
Chandra et al. [22]	- Image processing strategy used to segment the lung X-ray. - Five classifiers benchmark were used (i.e., DLP, Random forest, SMO, Logistic Regression, and Classification via Regression).	- Segmented lungs region and Region of Interest confined feature extraction were used to overcome false-positive problems. - Highest accuracy achieved was 95.63% with the Logistic Regression classifier and 95.39% with DLP.	Limited Dataset (i.e., only 412 X-ray images in total); hence, despite the high accuracy achieved, the method cannot be generalized.
Ibrahim et al. [23]	The authors used the DL approach based on a pre-trained AlexNet model to classify COVID-19, non-COVID-19 viral pneumonia, bacterial pneumonia, and normal CXR.	A four-way classification was used (i.e., three types of pneumonia and normal CXR images), whereas the current state-of-the-art usually addresses three-way classification, which achieved an accuracy of 93.42%, a sensitivity of 89.18%, and a specificity of 98.92%.	Limited Dataset (i.e., only 371 images of pneumonia caused by COVID-19 compared with over 4000 ones per each type of pneumonia and almost 3000 healthy lungs images); hence, the method cannot be generalized.
Tuncer et al. [24]	The authors used fuzzy tree processing on X-ray pictures. The novelty relies on the DL model used, named the 'exemplar model'.	- The production of three different feature pictures enhances model performance. - The best classifier obtained was the Cubic SVM which achieved a 97.01% classification accuracy.	Limited Dataset (i.e., 435 chest x-ray images divided into 135 COVID-19, 150 pneumonia patients, and 150 healthy lungs); hence the method cannot be generalized.
Stephen et al. [25]	The authors created a seven-layer CNN model for X-ray picture classification and used automatic feature learning.	The performances obtained were: 13% of training loss, 95% of training accuracy, 18% of validation loss, and 94% of validation accuracy.	The main limitation of this study is due to the depth of data.
Sharma et al. [26]	The authors created a deep CNN model for X-ray picture classification. For more complicated tasks, automatic feature learning was used.	- The deep CNN model helps to extract features, which were used for image classification to identify whether an individual has pneumonia, achieving faster and more accurate results compared to manual techniques, from chest X-ray images. - Testing accuracy: 90.68%, with their model having both data augmentation and dropout.	Imbalanced dataset (i.e., 5863 images of chest X-ray of which around 4000 of them were of lungs affected by pneumonia and only around 1600 images were of healthy lungs).
Jaiswal et al. [27]	The authors developed a Mask Region-based CNN (Mask R-CNN) for segmentation jointly with thresholds in the background while training to obtain accurate results without using larger images.	Their proposed approach achieved better performances evaluated on the chest radiography dataset, which depicts potential pneumonia causes. Their approach avoids a prohibitive computational cost that would otherwise have occurred.	Imbalanced dataset; i.e., there was an abundance of negative instances (healthy cases) and a scarcity of positive instances (cases of the disease).

[19] discussed the challenges and the future of medical image processing. According to Dinggang Shen [20], much work has been done on applying DL-based algorithms to diagnose various illnesses. In 1989, LeCun et al. [21] utilized CNNs for the first time to recognize handwritten zip codes. We employed CNN models because of their remarkable advantages in detecting significant characteristics from a given input image, without requiring any human intervention.

3.2. Prominent pneumonia detection works

According to Liu et al. study [28], pneumonia was responsible for approximately 15.6% of the 5.9 million deaths of children under five in 2015. Prompt diagnoses and treatments significantly reduce this mortality rate. Conversely, the contrast in chest X-ray pictures [29], on the other hand, is poor, rendering manual examination ineffective [30].

The literature is rich in research on automated pneumonia diagnosis and detection using X-ray images. For our evaluation, we selected representative related works based on deep learning methods and conventional neural networks for pneumonia classification. We study their approaches, datasets, advantages, and shortcomings, individually. We then replicate those methods under a uniform data science pipeline, to make insights based on a fair comparison. Table 1 provides a detailed comparison of the related work in detecting pneumonia, as discussed next.

Computer-assisted diagnosis [31] has the potential to improve efficiency and speed up therapy. For example, Chandra et al. [22] separated lung areas from chest X-ray images and retrieved eight statistical parameters to identify them. Ibrahim et al. [23] employed six characteristics to diagnose pneumonia in 52 patients' chest CT scan pictures;

the best region under the receiver operating characteristic was that they could get a 97% curve value. However, in both those works, the methods cannot be generalized as the study was conducted on a limited dataset.

Tuncer et al. [24] employed an approach based on DL. They started by transforming the images with a fuzzy tree transformation. After retrieving features using a multi-kernel local binary pattern, the data were categorized using standard classifiers. Finally, they tested the approach on a dataset of minor pneumonia and COVID-19 samples, finding it to be 97.01% accurate. The main limitation of this study, is its reliance on a small-size dataset.

Stephen et al. [25] created CNN architectures to categorize pneumonia chest X-ray pictures, achieving an overall and an accuracy rate of 93.73% was achieved. To compensate for the shortage of data, they employed data augmentation, leading to a slight improvement in performance.

In line with the work of Stephen et al. Sharma et al. [26] also developed a similar model, achieving an overall accuracy of 90.68%. In spite of employing dropout and data augmentation, their model still suffered from a limited as well as imbalanced dataset.

To detect pneumonia traces, Jaiswal et al. [27] employed a region-based CNN, using an ensemble model consisting of ResNet-101 and ResNet-50 for image thresholding. The primary limitation derived, again, from having imbalanced classes.

Limited data availability, which is typical in medical datasets, is often addressed via data augmentation, to increase the amount of data from which the CNNs can learn, as reported in the work by Rajpukar et al. [29], that could only achieve a 76.8% F1-score.

Herein, to overcome data limitations, we are instead using a larger dataset, proceeding then with its curation, class balancing and data

augmentation. We also specialized the six classifiers under scrutiny in order to distinguish between conventional pneumonia (bacterial or viral) from COVID-19.

3.3. The COVID-19 works

COVID-19, classified as a pandemic by the World Health Organization, has drastically impacted people's everyday lives, immediate health, and economy worldwide [32]. Ozturk et al. [33] established a model for automatic COVID-19 identification using chest X-ray pictures. Their model provides accurate diagnoses for Binary Classification (no findings vs. COVID) and multi-class Classification (Pneumonia, COVID-19, and Normal), and the former was 98.08% accurate.

The general relevance of AI approaches in image analysis for recognizing and controlling COVID-19 instances was described by researchers in [34]. In addition, DL algorithms for detecting COVID-19 was discussed in [34]. Researchers in [35] created a COVID-19 diagnostic method based on CNNs.

Another relevant work reported an X-ray dataset containing pictures of typical COVID-19 patients, general pneumonia patients, and those with no illness [36]. The study used cutting-edge CNN architectures to recognize COVID-19 patients automatically. In this work, transfer learning was adopted to achieve a 97.82% in COVID-19 identification.

Another recent significant approach aimed at validating and adapting deep CNNs is the Decompose, Transfer, and Compose (*i.e.*, DeTraC) one [37], which is a novel CNN architecture based on class decomposition [38]. The authors achieved an accuracy of 95.12%, a sensitivity of 97.91%, and a specificity of 91.87%.

Several recent studies have explored the application of deep learning models for classifying chest X-ray images to diagnose COVID-19 and pneumonia. For instance, Gielczyk et al. [7] focused on the impact of pre-processing techniques, achieving high F1-scores by improving data pre-processing methods such as thresholding and blurring. Islam et al. [8] applied a combined CNN-RNN architecture with transfer learning for the diagnosis of COVID-19, achieving promising results through VGG19-RNN. Another study by Saad et al. [9] optimized CNN hyperparameters like learning rates and momentum using grid search, which significantly improved classification performance for COVID-19 using X-ray and CT images. Balaha et al. [10] developed a hybrid framework integrating deep learning with genetic algorithms to optimize hyperparameters, achieving state-of-the-art results using multiple CNN architectures, including VGG16 and ResNet-50.

While these studies have made substantial contributions, our work distinguishes itself by utilizing a large and balanced dataset of 6939 X-ray images, specifically covering both common pneumonia and COVID-19 pneumonia. Our comparison of multiple CNN architectures (VGGNet, ResNet, AlexNet, SqueezeNet, DenseNet, and InceptionV3) with pre-trained weights provides a comprehensive, fair evaluation of model performance. Importantly, we demonstrate the superiority of the VGGNet architecture for this task, achieving an overall F1-score accuracy of 97%. Moreover, our study combines the same data augmentation and normalization techniques in all six to address class imbalance and enhance model robustness to a level which is applicable to real-world clinical scenarios.

4. Methodology and proposed solution

The created system's architecture is shown in Fig. 3, which contains the following steps: (i) importing the dataset composed by three selected classes, (ii) pre-processing the data, (iii) extracting and classifying the features through CNNs, and (iv) obtaining the output classifier. We recall that the three selected classes are: (i) general pneumonia (both bacterial or viral) labeled as 'Pneumonia', (ii) the specific case of pneumonia caused by COVID-19 labeled as 'COVID-19', and (iii) the images without any pneumonia (nor other diseases) labeled as 'Normal'.

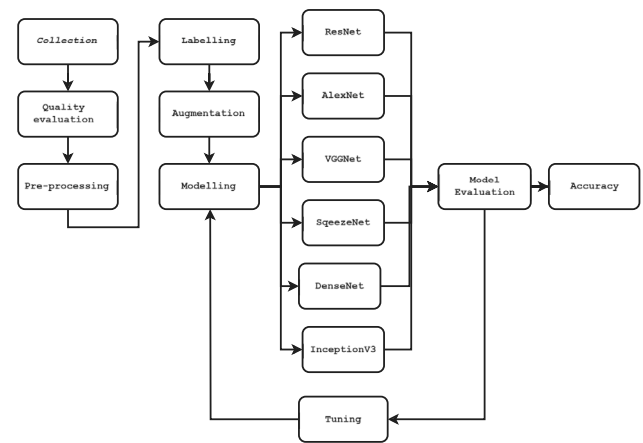


Fig. 3. Data Science Pipeline. The image shows all the steps followed to perform the analyses from the data collection up to final accuracy of the model.

Our pipeline comprises six pre-trained models, an image data generator, and flattened, dense, and dropout layers. Before identifying pneumonia and COVID-19, we pre-processed the data and perform data augmentation (See Section 4.1). A classification matrix computes: precision, recall, F1 score, and overall accuracy.

4.1. The dataset

We used the publicly available dataset from the Kaggle dataset composed of 6939 X-ray images. The X-ray images dataset was chosen for this study due to its comprehensive coverage of diverse pathologies, including common bacterial and viral pneumonia, COVID-19-induced pneumonia, and normal chest X-rays. With 6939 images, the dataset is large and well-balanced across the three classes, which helps ensure that the training and testing of models are robust and unbiased.

The real-world clinical dataset is another key factor, as chest X-rays are widely used, affordable, and quick diagnostic tools in healthcare, making the findings of this study directly applicable to practical medical settings. As said, the dataset is publicly available on Kaggle, promoting transparency and reproducibility, and it comes preprocessed, which facilitates the application of deep learning models without requiring extensive pre-processing work. In addition we had the dataset screened by an expert radiologist, to remove all unreadable or low-quality scans.

The dataset was, then, divided into training samples (80%) and testing samples (the remaining 20%). Each set included samples from all three classes (*i.e.*, 2.313 cases for each class): Pneumonia, COVID-19, and Normal. The Pneumonia class contained both bacterial and viral pneumonia cases, making an exception for pneumonia caused by COVID-19 viral that was, instead, labeled separately.

In order to make the dataset ready for training, we pre-processed the dataset as described later in this section. To improve generalization and prevent overfitting, we used flatten, dense, and dropout layers in our CNN models. Additionally, we used data augmentation and an image data generator to balance class distributions during training.

Data augmentation. As said, due to the limited number of available chest X-ray images and the classes imbalance, we applied data augmentation techniques to synthetically expand the dataset and reduce the risk of overfitting.

Fig. 4 illustrates examples of the augmented chest X-ray images. The augmentations applied during training and included both *pixel-level* and *feature-level* transformations are the following:

- **Normalization:** Pixel values were scaled to the [0,1] range to guarantee faster convergence and improve training stability.

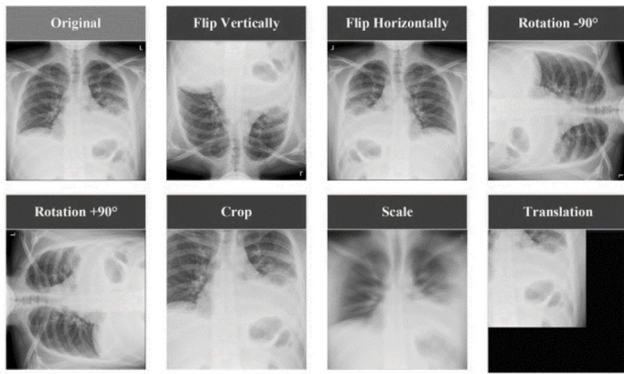


Fig. 4. The figure shows different sample images to highlight the data augmentation techniques used to balance the dataset: flipping, rotations, cropping, scaling and translating.

- *Color space augmentations:* Brightness and contrast adjustments were applied to simulate varying imaging conditions.
- *Noise handling:* Kernel-based filters were used to introduce noise, improving the models' robustness to low-quality images.
- *Image resizing:* All input images were resized to a uniform size of $224 \times 224 \times 3$ to match the input dimensions of the CNN architectures used.
- *Horizontal flipping and rotation:* Random affine transformations (e.g., scaling, translation, and rotation) were applied while preserving the spatial integrity of key anatomical features. Images were randomly flipped horizontally and rotated within a 15° range to increase variation in image orientation.
- *Random Erasing,* which a technique that removes small random patches from an image to force the model to generalize across different regions.

Note that each image was also converted from a PIL.Image or NumPy.ndarray format into a PyTorch tensor (e.g., ToTensor transform) for compatibility with the training pipeline.

4.2. The convolutional neural networks

As part of the ablation studies, we compared six Convolutional Neural Network (CNN) architectures, namely (i) *ResNet*, (ii) *AlexNet*, (iii) *VGGNet*, (iv) *SqueezeNet*, (v) *DenseNet*, and (vi) *InceptionV3*.

These models, indeed, show good performance in handling image classification tasks especially within the medical imaging domain. Characterized by their unique approaches and capabilities towards feature extraction, every architecture is able to go a step further.

More specifically, *ResNet* aims at constructing deep networks, and uses skip connections to remove learning bottlenecks and increase convergence, that may be caused by the depth of architecture. It uses residual connections, which helps mitigate the degradation problem in deep networks, that allow gradients to flow through the network without vanishing, enabling the training of very deep networks (up to 152 layers).

AlexNet consists of 5 convolutional layers followed by 3 fully connected layers. Its importance relies on the fact that it was one of the first CNNs to demonstrate the effectiveness of deep learning for image classification tasks, winning the ImageNet competition in 2012 [19].

VGGNet emphasizes depth and simplicity with its homogeneous architecture, which helps in feature extraction. It uses a very deep architecture (up to 19 layers) with small 3×3 convolutional filters and a uniform architecture across the network.

SqueezeNet uses a fire module that compresses the input using 1×1 convolutions followed by 3×3 convolutions, reducing the number of parameters while maintaining accuracy. Its lightweight design allows it

to achieve competitive performance with a significantly smaller model size. However, in medical imaging domain, this method is not expected to outperform the other models under scrutiny.

DenseNet connects each layer to every other layer in a feed-forward fashion, with each layer receiving inputs from all preceding layers and passing its own feature maps to all subsequent layers. This model promotes feature reuse and reduces the number of parameters compared to traditional architectures. The main strength of this model is that *DenseNet* improves gradient flow and feature propagation, leading to better performance with fewer parameters, which is useful in scenarios with limited data.

Lastly, the *Inception* family of models (e.g., *InceptionV1*, *InceptionV2*, and *InceptionV3*) uses a module-based architecture that combines multiple filter sizes at each layer (1×1 , 3×3 , and 5×5 convolutions) to capture features at various scales. The key feature is that they significantly improve performance on image classification tasks, while reducing the number of parameters, making it more efficient than the other models.

Transfer learning was employed in consideration of the fact that there are image dataset models such as labeled images. ImageNet weight was used in order to improve performance on the relatively small data set we had.

4.3. The fine-tuning

Deep CNN models trained on large, open datasets using pre-trained networks can be effective for classifying general images. However, without further fine-tuning, they are not suitable for specific medical image classification tasks. Indeed, training a deep CNNs from scratch is challenging due to the need for extensive labeled data and expertise to achieve proper convergence. An effective alternative is to fine-tune a pre-trained CNN, which has already been trained on a large set of labeled images that have been shown to be effective in the context of medical image analysis [39]. The fine-tuning process involves adapting a pre-trained model for specific tasks or datasets by transferring its weights to the new network.

Thus, we started from a model that had already been pre-trained on a public image dataset. Then, we tuned it by adjusting the model weights through transfer learning, for each of the six deep CNNs under scrutiny (i.e., *ResNet*, *AlexNet*, *VGGNet*, *SqueezeNet*, *DenseNet*, and *InceptionV3*). Next, we unfroze some final blocks and updated their weights at each training phase. Unfreezing (resp., freezing) a layer means that it will be allowed (resp., not allowed) to update their weights during the fine-tuning process. We should note that fine-tuning requires changing the CNN architecture and re-training the network to learn new object types — this is a highly computationally-intensive process requiring significant time.

Once the architecture had been adjusted and the layers to be frozen (or unfrozen) had been chosen, we could train the models for our classification task. The details of the training are described in Section 4.5.

4.4. Experimental setup

In this section we provide further details on the experimental setup.

Number of experiments. As regard variability in model performance, each model was trained and validated multiple times (approximately 10 runs per model), so as to ensure reproducibility and reliability and consistency of results across different CNN's architectures. This means, there was a total of 60+ main individual experiments.

Train and validation sets split. The dataset was divided at a ratio of 80% for training and 20% for validation. The training data was essential for fitting the model while the validation set was critical in checking how the fitted model would perform in piece of data that it has not seen before.

Table 2

Models performances for overall Accuracy, Precision, Recall, and F1-Score per each of the three classes (i.e., Pneumonia, COVID-19, and healthy lungs).

Model	Accuracy (%)	Precision (%)			Recall (%)			F1-Score (%)		
		Pneumonia	COVID-19	Normal	Pneumonia	COVID-19	Normal	Pneumonia	COVID-19	Normal
ResNet	95	92	90	96	91	88	95	91	89	97
AlexNet	90	88	85	92	87	84	91	87	84	91
VGG-19	97	96	91	99	98	97	88	97	93	94
SqueezeNet	88	85	82	90	83	80	89	84	81	89
DenseNet	92	89	87	95	88	85	94	88	86	95
InceptionV3	94	91	89	95	90	86	96	90	87	95

Cross-validation. We performed a 5-fold cross validation to further validate the models in terms of robustness, stability, and data independence.

4.5. The training

Each model was subjected to the following training pipeline:

Batch size. We tested out several batch sizes such as 16, 32, 64 in a bid to improve GPU memory usage efficiency as well as the training.

Epochs. We performed several training sessions and altered the epochs in every run (averagely 20–30 epochs per model) to check model convergence.

Optimizer and learning rate. We applied *adam optimizer* at 0.001 as an initial learning rate in the beginning stages of training, the learning rate was then altered for tuning and validation purposes.

Loss function. The categorical cross entropy loss function was utilized as this is suited to multiclass classification tasks.

Dropout and batch normalization. So as not to have cases of overfitting which would affect models since they would perform perfectly on models but poorly on new data, dropout layers and batch normalizations were incorporated into the models.

We divided the 6939 input photos (achieved as the result of a selection and augmentation process) between the training set (including 80% of the overall data) and the testing set (the remaining 20% of data). The chosen testing set was independent of the training folder and was used to assess training progress. After the training process, the model trained was evaluated by additional independent tests. Dropout is a strategy we used for preventing overfitting applied to the last fully connected layers. Early halting was employed by keeping track of the testing loss at each epoch. Finally, the best model was found by examining the test dataset after deep CNN model training. After the training stage, the accuracy and precision–recall (PR) were calculated to evaluate the test datasets' performance thoroughly.

5. Experimental results

In this section, the experimental results are discussed.

Following the training and validation of all models, it was found that VGGNet was the best model with an F1-score accuracy of about 97% on common pneumonia disease classification, 93% on COVID-19 pneumonia and 94% in healthy lungs. Having identified the optimal performing model, we proceeded to evaluate its performance on a test set comprising 20% of the dataset, which had not been utilized during the training and validation phases. Additionally, we employed Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the specific regions of the chest X-rays that contributed to the model's predictions. This analysis confirmed that the model focused on relevant features, such as lung opacities and other abnormalities, rather than learning spurious correlations or irrelevant information.

It is important to recall that we considered VGG-19, which is a version of VGGNet with 19 weight layers (16 convolutional layers and 3

Table 3

The confusion matrix displaying the performance of a classification model. The matrix compares the three predicted class labels (i.e., healthy lungs or normal, common pneumonia and pneumonia caused by COVID-19) with actual class labels.

		Predicted		
		Normal	Pneumonia	COVID-19
Actual	Normal	444	2	9
	Pneumonia	6	408	48
	COVID-19	14	1	448

fully connected layers). Indeed, deeper models, such as VGG-19, tend to perform better on image classification tasks but at the cost of increased computational complexity and memory usage.

Once the best CNN model was selected (i.e., VGGNet or, more specifically, VGG-19) and after the normalization and augmentation techniques, we moved on to the next step, the training of the model. The parameters we pass to the model are train sets, the overall epochs (that we set equal to 5), and the validation set used to calculate Val-loss and Val-accuracy. Again, the best training and action accuracy were confirmed to be achieved by the VGGNet CNN model. Table 2 provides a detailed summary of the performances achieved by the six models under scrutiny.

VGG-19, as said, is the best model with an overall 97% accuracy, and with a precision of 96% for pneumonia, 91% for COVID-19 and 99% for normal (e.g., healthy) cases. We recall that this high precision indicates that when the model predicts a positive class, it is likely correct. It is particularly important in scenarios where false positives are costly or undesirable, such as in medical diagnoses where a false positive could lead to unnecessary treatment.

For the same model, the recall, which is the model sensitivity or true positive rate, is of 98% for pneumonia and 97% for COVID-19 and 88% for normal cases. For a fair comparison, we also computed the F1-score, which computes the average of precision and recall. In VGG-19 we achieved 97% for pneumonia, 93% for Covid 19 and 94% for normal conditions.

On the other hand, DenseNet and AlexNet have the worst overall accuracy of 92% and 90%, respectively., though DenseNet performs a little better in terms of pneumonia precision at 89%. SqueezeNet though recording the lowest accuracy level of 88% still manages to correctly predict and recall pneumonia, Covid and normal cases even though it has quite low predictive values.

Table 3 shows the confusion matrix related to the same three classes (pneumonia, COVID-19, and normal) for VGGNet. This confirms that it is more difficult to differentiate between the origin of pneumonia (i.e., whether or not it is COVID-19 dependent) than the existence of the disease itself. However, the F1-score for pneumonia of 97% indicates that it is, indeed, possible to differentiate pneumonia causes up to a good extent, whilst providing an overall 94% ability to well differentiate between pathological and non-pathological cases.

6. Conclusions

In this work, we established a rigorous data science pipeline, from data ingestion to model validation, to conduct a fair comparison among prominent CNN models (*i.e.*, *ResNet*, *AlexNet*, *VGGNet*, *SqueezeNet*, *DenseNet*, and *InceptionV3*) to address medical challenges. This was necessary because the numerous studies in the literature are hardly comparable, due to a variety of differences in their data science pipelines.

To sum up, we used chest X-ray images from a significantly larger and well-balanced dataset than the ones found in the literature, aiming to distinguish between common pneumonia (both viral and bacterial) and the one caused by COVID-19, to conduct a proper screening. Indeed, the use of automatic chest X-ray classification using CNNs provides significant advantages, mainly when using several cascading CNN convolutions.

Our approach allowed us to identify *VGGNet* as the most accurate model for the specific classification problem at hand. One of the most significant results is that our work also addressed the research question regarding the necessary dataset size to achieve good results (*i.e.*, 94% accuracy).

Thanks to our methodical comparative evaluation we have also been able to answer another crucial research question relating to the order of magnitude required for such medical datasets, finding that about 7000 samples provide a good starting point for medical grade classification. It was to be expected that, in order to increase the number of classes of the classifier, a larger dataset would be required. In fact, our models were not effectively able to differentiate among the more specific types of pneumonia — we had to aggregate the conventional pneumonia types (viral and bacterial) in order to achieve good discrimination in the detection of COVID-19. However, our overall study indicates that, with further data samples, it will be possible to increase the classification discriminative ability.

For future works, thus, there is a need to improve the tertiary classifier (*i.e.*, distinguishing between non-pathological, generic pneumonia, and COVID-19 pneumonia). The key to tackling this problem lies in a combination of high-quality data, more elaborate data augmentation, and pre-processing techniques. The aim is also to extend this approach to other types of image-based pathologies and compare the results with those.

CRedit authorship contribution statement

Hanan Aljuaid: Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Hanan Adlan:** Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bu-door Alkebsi:** Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Badria Sulaiman Alfurhood:** Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Antonio Liotta:** Writing – review & editing, Supervision. **Lucia Cavallaro:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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